

Nazwa Aulia Dwi Purnomo - 5027251018

Putri Permata Sabila - 5027251047

## **CONFIG**

### **Router(Lain)**

auto eth0

iface eth0 inet dhcp

auto eth1

iface eth1 inet static

address 192.229.1.1

netmask 255.255.255.0

auto eth2

iface eth2 inet static

address 192.229.2.1

netmask 255.255.255.0

auto eth3

iface eth3 inet static

address 192.229.3.1

netmask 255.255.255.0

### **Alice**

auto eth0

iface eth0 inet static

address 192.229.1.2

netmask 255.255.255.0

gateway 192.229.1.1

### **Mika**

```
auto eth0
iface eth0 inet static
    address 192.229.1.3
    netmask 255.255.255.0
    gateway 192.229.1.1
```

### **Chisa**

```
auto eth0
iface eth0 inet static
    address 192.229.2.2
    netmask 255.255.255.0
    gateway 192.229.2.1
```

### **Knights**

```
auto eth0
iface eth0 inet static
    address 192.229.3.2
    netmask 255.255.255.0
    gateway 192.229.3.1
```

### **Eiri**

```
auto eth0
iface eth0 inet static
    address 192.229.3.3
    netmask 255.255.255.0
    gateway 192.229.3.1
```

## **Masukkan script otomatis**

```
Eiri:~# cat /etc/network/interfaces
```

```
auto eth0
```

```
iface eth0 inet static
```

```
    address 192.229.3.3
```

```
    netmask 255.255.255.0
```

```
    gateway 192.229.3.1
```

```
up echo "nameserver 8.8.8.8" > /etc/resolv.conf
```

```
root@Lain:~# cat init.sh
```

```
/root/router.sh
```

```
root@Lain:~# cat router.sh
```

```
iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
```

```
iptables -A FORWARD -i eth1 -o eth0 -j ACCEPT
```

```
iptables -A FORWARD -i eth2 -o eth0 -j ACCEPT
```

```
iptables -A FORWARD -i eth3 -o eth0 -j ACCEPT
```

## Nomor 7 BOS

Apk update

apk add vsftpd

Vsftpd -v

echo -e "alice\nmika" > /etc/vsftpd.userlist

mkdir -p /etc/vsftpd/user\_conf

echo "write\_enable=NO" > /etc/vsftpd/user\_conf/mika

nano /etc/vsftpd.conf

Isinya:

Chisa:~# cat /etc/vsftpd.conf

listen=YES

anonymous\_enable=NO

local\_enable=YES

write\_enable=YES

local\_root=/var/wired/data

userlist\_enable=YES

userlist\_file=/etc/vsftpd.userlist

userlist\_deny=NO

user\_config\_dir=/etc/vsftpd/user\_conf

seccomp\_sandbox=NO

pam\_service\_name=vsftpd

pasv\_enable=YES

pasv\_min\_port=30000

pasv\_max\_port=30100

file\_open\_mode=0666

local\_umask=002

Chisa:~#

## INI BUAT JALANIN 3 USERNYA

vsftpd /etc/vsftpd.conf &

## CARA TESNYA

### lftp 192.229.2.2

Masukin usernya

User alice, pass: alice1234

User mika, pass: mika1234

User eiri, pass: eiri1234

Seharusnya

alice -> bisa put, get, ls,

mika-> bisa get, g bisa put. Karena dia read only

```
lftp mika@192.229.2.2:~> ls
-rw-rw-r-- 1 1000 1003          16 Sep 15 20:12 signal_alice.txt
lftp mika@192.229.2.2:~> get signal_alice.txt
16 bytes transferred
lftp mika@192.229.2.2:~> put signal_alice.txt
put: Access failed: 550 Permission denied. (signal_alice.txt)
lftp mika@192.229.2.2:~> █
```

eiri-> ga bisa ls, semua g bisa, karena dia ga bisa login

Trus ini jujur gatau kutambah2in karena error, soalnya si mika kok ga bisa get

chown alice:ftpaccess /var/wired/data/signal\_alice.txt

chmod 664 /var/wired/data/signal\_alice.txt

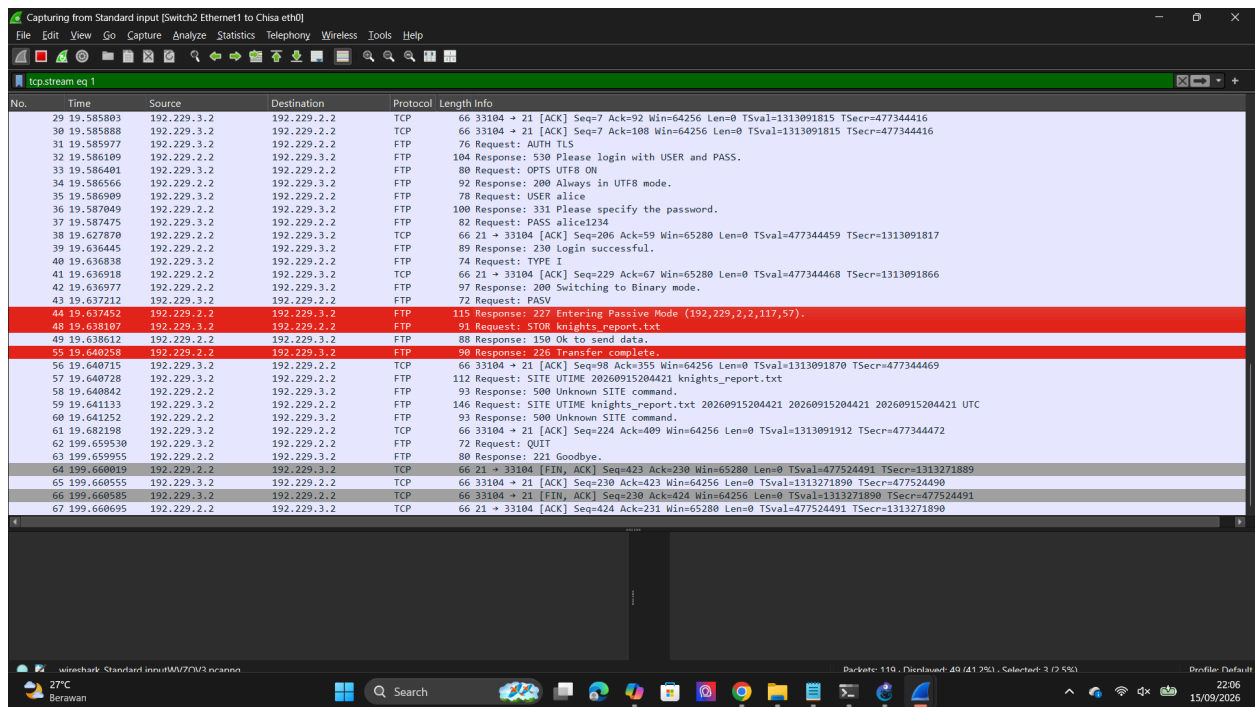
## 8. Buat link drivenya itu pindahkan ke cisca

Nano knights\_report.txt

Copas kedalem

Trs masuk ke user alice

Put knights\_report.txt



9. **cd /var/wired/data/**

Nano protocol7\_manifesto.txt, copas aja dari drive

Get protocol7\_manifesto.txt

Put protocol7\_manifesto.txt

```
lftp mika@192.229.2.2:~> ls
-rw-rw-r-- 1 1000 1000      1111 Sep 15 20:49 knights_report.txt
-rw-r--r-- 1 0 0          1738 Sep 15 21:18 protocol7_manifesto.txt
-rw-rw-r-- 1 1000 1003      16 Sep 15 20:12 signal_alice.txt
lftp mika@192.229.2.2:~> get protocol7_manifesto.txt
1738 bytes transferred
lftp mika@192.229.2.2:~> put protocol7_manifesto.txt
put: Access failed: 550 Permission denied. (protocol7_manifesto.txt)
lftp mika@192.229.2.2:~> █
```

## 10. ping -c 77 -s 128 -i 0.3 192.229.2.2

| No. | icmp     | Source      | Destination | Protocol | Length | Info                                                             |
|-----|----------|-------------|-------------|----------|--------|------------------------------------------------------------------|
| 1   | icmpv6   | 192.229.3.2 | 192.229.2.2 | ICMP     | 170    | Echo (ping) request id=0x0421, seq=1/256, ttl=63 (reply in 2)    |
| 2   | 0.000189 | 192.229.2.2 | 192.229.3.2 | ICMP     | 170    | Echo (ping) reply id=0x0421, seq=1/256, ttl=64 (request in 1)    |
| 3   | 0.304046 | 192.229.3.2 | 192.229.2.2 | ICMP     | 170    | Echo (ping) request id=0x0421, seq=2/512, ttl=63 (reply in 4)    |
| 4   | 0.304232 | 192.229.2.2 | 192.229.3.2 | ICMP     | 170    | Echo (ping) reply id=0x0421, seq=2/512, ttl=64 (request in 3)    |
| 5   | 0.607992 | 192.229.3.2 | 192.229.2.2 | ICMP     | 170    | Echo (ping) request id=0x0421, seq=3/768, ttl=63 (reply in 6)    |
| 6   | 0.608177 | 192.229.2.2 | 192.229.3.2 | ICMP     | 170    | Echo (ping) reply id=0x0421, seq=3/768, ttl=64 (request in 5)    |
| 7   | 0.912079 | 192.229.3.2 | 192.229.2.2 | ICMP     | 170    | Echo (ping) request id=0x0421, seq=4/1024, ttl=63 (reply in 8)   |
| 8   | 0.912294 | 192.229.2.2 | 192.229.3.2 | ICMP     | 170    | Echo (ping) reply id=0x0421, seq=4/1024, ttl=64 (request in 7)   |
| 9   | 1.212285 | 192.229.3.2 | 192.229.2.2 | ICMP     | 170    | Echo (ping) request id=0x0421, seq=5/1280, ttl=63 (reply in 10)  |
| 10  | 1.212478 | 192.229.2.2 | 192.229.3.2 | ICMP     | 170    | Echo (ping) reply id=0x0421, seq=5/1280, ttl=64 (request in 9)   |
| 11  | 1.519850 | 192.229.3.2 | 192.229.2.2 | ICMP     | 170    | Echo (ping) request id=0x0421, seq=6/1536, ttl=63 (reply in 12)  |
| 12  | 1.519975 | 192.229.2.2 | 192.229.3.2 | ICMP     | 170    | Echo (ping) reply id=0x0421, seq=6/1536, ttl=64 (request in 11)  |
| 13  | 1.823880 | 192.229.3.2 | 192.229.2.2 | ICMP     | 170    | Echo (ping) request id=0x0421, seq=7/1792, ttl=63 (reply in 14)  |
| 14  | 1.824029 | 192.229.2.2 | 192.229.3.2 | ICMP     | 170    | Echo (ping) reply id=0x0421, seq=7/1792, ttl=64 (request in 13)  |
| 15  | 2.127937 | 192.229.3.2 | 192.229.2.2 | ICMP     | 170    | Echo (ping) request id=0x0421, seq=8/2048, ttl=63 (reply in 16)  |
| 16  | 2.128055 | 192.229.2.2 | 192.229.3.2 | ICMP     | 170    | Echo (ping) reply id=0x0421, seq=8/2048, ttl=64 (request in 15)  |
| 17  | 2.431982 | 192.229.3.2 | 192.229.2.2 | ICMP     | 170    | Echo (ping) request id=0x0421, seq=9/2304, ttl=63 (reply in 18)  |
| 18  | 2.432108 | 192.229.2.2 | 192.229.3.2 | ICMP     | 170    | Echo (ping) reply id=0x0421, seq=9/2304, ttl=64 (request in 17)  |
| 19  | 2.735906 | 192.229.3.2 | 192.229.2.2 | ICMP     | 170    | Echo (ping) request id=0x0421, seq=10/2560, ttl=63 (reply in 20) |
| 20  | 2.736015 | 192.229.2.2 | 192.229.3.2 | ICMP     | 170    | Echo (ping) reply id=0x0421, seq=10/2560, ttl=64 (request in 19) |
| 21  | 3.039866 | 192.229.3.2 | 192.229.2.2 | ICMP     | 170    | Echo (ping) request id=0x0421, seq=11/2816, ttl=63 (reply in 22) |
| 22  | 3.039988 | 192.229.2.2 | 192.229.3.2 | ICMP     | 170    | Echo (ping) reply id=0x0421, seq=11/2816, ttl=64 (request in 21) |
| 23  | 3.344253 | 192.229.3.2 | 192.229.2.2 | ICMP     | 170    | Echo (ping) request id=0x0421, seq=12/3072, ttl=63 (reply in 24) |
| 24  | 3.344372 | 192.229.2.2 | 192.229.3.2 | ICMP     | 170    | Echo (ping) reply id=0x0421, seq=12/3072, ttl=64 (request in 23) |
| 25  | 3.644581 | 192.229.3.2 | 192.229.2.2 | ICMP     | 170    | Echo (ping) request id=0x0421, seq=13/3328, ttl=63 (reply in 26) |
| 26  | 3.644736 | 192.229.2.2 | 192.229.3.2 | ICMP     | 170    | Echo (ping) reply id=0x0421, seq=13/3328, ttl=64 (request in 25) |
| 27  | 3.951846 | 192.229.3.2 | 192.229.2.2 | ICMP     | 170    | Echo (ping) request id=0x0421, seq=14/3584, ttl=63 (reply in 28) |
| 28  | 3.952069 | 192.229.2.2 | 192.229.3.2 | ICMP     | 170    | Echo (ping) reply id=0x0421, seq=14/3584, ttl=64 (request in 27) |
| 29  | 4.255922 | 192.229.3.2 | 192.229.2.2 | ICMP     | 170    | Echo (ping) request id=0x0421, seq=15/3840, ttl=63 (reply in 30) |
| 30  | 4.256041 | 192.229.2.2 | 192.229.3.2 | ICMP     | 170    | Echo (ping) reply id=0x0421, seq=15/3840, ttl=64 (request in 29) |
| 31  | 4.559955 | 192.229.3.2 | 192.229.2.2 | ICMP     | 170    | Echo (ping) request id=0x0421, seq=16/4096, ttl=63 (reply in 32) |

Frame 1: Packet, 170 bytes on wire (1360 bits), 170 bytes captured (1360 bits) on interface -, id 0

Ethernet II, Src: 02:42:71:43:9f:02 (02:42:71:43:9f:02), Dst: 02:42:d2:1d:cd:00 (02:42:d2:1d:cd:00)

Internet Protocol Version 4, Src: 192.229.3.2, Dst: 192.229.2.2

Internet Control Message Protocol

0000 02 42 d2 1d cd 00 02 42 71 43 9f 02 08 00 45 00 B...B qC...E

0010 00 9c c2 bc 40 00 3f 01 f1 d5 c0 e5 03 02 c0 e5 ...@.?.....

0020 02 02 08 00 b7 af 04 21 00 01 2d bf a9 6a 00 00 .....f.....

0030 00 00 c8 34 05 00 00 00 00 10 11 12 13 14 15 ...4.....

0040 16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25 .....!\*%\$

0050 26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35 &'()\*+,-./012345

0060 36 37 38 39 3a 3b 3c 3d 3e 3f 40 41 42 43 44 45 6789:;<=>?@ABCDE

0070 46 47 48 49 4a 4b 4c 4d 4e 4f 50 51 52 53 54 55 FGHIJKLM NOPQRSTU

0080 56 57 58 59 5a 5b 5c 5d 5e 5f 60 61 62 63 64 65 VWXYZ[\]^\_abcde

0090 66 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 fg hijklm nopqrstu

## Analysis

```
--- 192.229.2.2 ping statistics ---
77 packets transmitted, 77 received, 0% packet loss, time 25671ms
rtt min/avg/max/mdev = 0.409/0.622/1.060/0.128 ms
Knights:~#
```

## 11 yaAllah

Konsol Chisa : apk update

apk add busybox-extras

apk add shadow # biar ada useradd/chpasswd versi lengkap

useradd -m phantom\_user

echo "phantom\_user:wired\_ghost" | chpasswd

Telnetd

Cek apa udh listen : netstat -tuln | grep 23

Konsol Eiri : apk update

apk add busybox-extras

which telnet

Cek konek sm chisa : ping -c 3 192.229.2.2

Capture ya

Konsol Eiri : telnet 192.229.2.2

Filter : tcp.port == 23

Hasil plain text

```
.....
.....K.!.....
Chisa login:
phantom_user
phantom_user

Password:
wired_ghost

Welcome to Alpine!

The Alpine Wiki contains a large amount of how-to guides and general
information about administrating Alpine systems.
See <https://wiki.alpinelinux.org/>.

You can setup the system with the command: setup-alpine

You may change this message by editing /etc/motd.

      .o.      oooo      o8o  ooooo  ooo      .
     .888.    `888      `""  `888b.  `8'      .o8
    .8"888.    888  oo,oooo.  oooo  8 `88b.   8  .ooooo. .o888oo
   .8' `888.   888  888' `88b `888  8 `88b.  8  d88' `88b  888
  .88ooo8888.  888  888  888  888  8 `88b.8  888ooo888  888
 .8' `888.   888  888  888  888  8 `888  888  .o  888 .
o88o   o8888o o888o 888bod8P' o888o o8o   `8 `Y8bod8P' "888"
      888
     o888o

AlpiNet - Lightweight Alpine-based Networking Toolbox
Type "alpinet-tools" for available utilities
Chisa:~$ .[6n
.[33;10R
```

**Simpan** : nano /root/setup\_telnet.sh

```
#!/bin/sh
```

```
apk update
```

```
apk add busybox-extras shadow
```

```
id phantom_user >/dev/null 2>&1 || useradd -m phantom_user
```

```
echo "phantom_user:wired_ghost" | chpasswd
```

```
telnetd
```

Lalu

```
chmod +x /root/setup_telnet.sh
```

```
echo "/root/setup_telnet.sh" >> /root/.bashrc
```

## **12 yaAllah**

Konsol Knights : apk update

```
apk add netcat-openbsd
```

```
nc -lk -p 22 &
```

```
nc -lk -p 80 &
```

```
netstat -tuln | grep -E '22|80' (pastiin dah listen smwa)
```

Konsol Alice : apk update

```
apk add netcat-openbsd
```

```
Cek konek : ping -c 3 <IP_Knights>
```

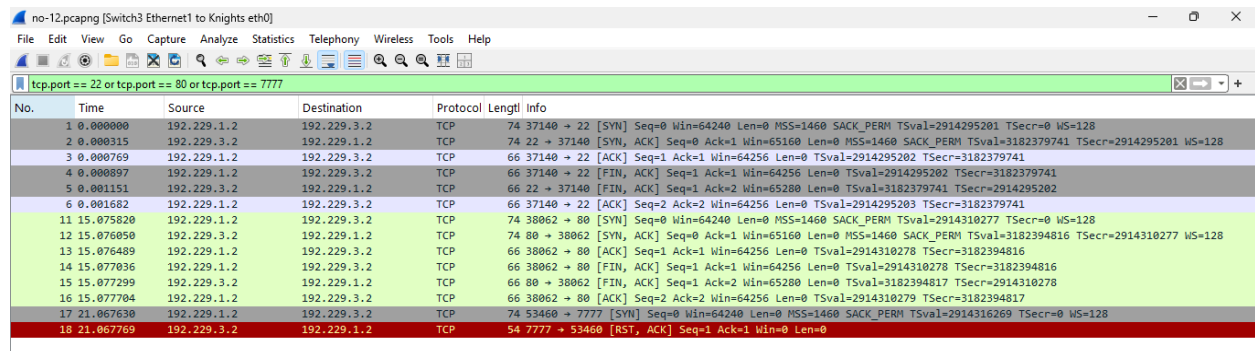
Capture ya

```
nc -vz <IP_Knights> 22
nc -vz <IP_Knights> 80
nc -vz <IP_Knights> 7777 (refused)
```

Filter : tcp.port == 22 or tcp.port == 80 or tcp.port == 7777

Analisis

```
Alice:~# nc -vz 192.229.3.2 22
Connection to 192.229.3.2 22 port [tcp/ssh] succeeded!
Alice:~# nc -vz 192.229.3.2 80
Connection to 192.229.3.2 80 port [tcp/http] succeeded!
Alice:~# nc -vz 192.229.3.2 7777
nc: connect to 192.229.3.2 port 7777 (tcp) failed: Connection refused
Alice:~#
```



The image shows a Wireshark packet capture window with the filter 'tcp.port == 22 or tcp.port == 80 or tcp.port == 7777'. The packet list shows several TCP connections. The first connection (No. 1) is a SYN packet from 192.229.1.2 to 192.229.3.2 on port 22. The second connection (No. 2) is a SYN-ACK packet from 192.229.3.2 to 192.229.1.2 on port 22. The third connection (No. 3) is an ACK packet from 192.229.1.2 to 192.229.3.2 on port 22. The fourth connection (No. 4) is a FIN packet from 192.229.1.2 to 192.229.3.2 on port 22. The fifth connection (No. 5) is a FIN-ACK packet from 192.229.3.2 to 192.229.1.2 on port 22. The sixth connection (No. 6) is an ACK packet from 192.229.1.2 to 192.229.3.2 on port 22. The seventh connection (No. 11) is a SYN packet from 192.229.1.2 to 192.229.3.2 on port 80. The eighth connection (No. 12) is a SYN-ACK packet from 192.229.3.2 to 192.229.1.2 on port 80. The ninth connection (No. 13) is an ACK packet from 192.229.1.2 to 192.229.3.2 on port 80. The tenth connection (No. 14) is a FIN packet from 192.229.1.2 to 192.229.3.2 on port 80. The eleventh connection (No. 15) is a FIN-ACK packet from 192.229.3.2 to 192.229.1.2 on port 80. The twelfth connection (No. 16) is an ACK packet from 192.229.1.2 to 192.229.3.2 on port 80. The thirteenth connection (No. 17) is a SYN packet from 192.229.1.2 to 192.229.3.2 on port 7777. The fourteenth connection (No. 18) is a RST-ACK packet from 192.229.3.2 to 192.229.1.2 on port 7777.

| No. | Time      | Source      | Destination | Protocol | Length | Info                                                                                                          |
|-----|-----------|-------------|-------------|----------|--------|---------------------------------------------------------------------------------------------------------------|
| 1   | 0.000000  | 192.229.1.2 | 192.229.3.2 | TCP      | 74     | 37140 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2914295201 TSecr=0 WS=128                     |
| 2   | 0.000315  | 192.229.3.2 | 192.229.1.2 | TCP      | 74     | 22 → 37140 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3182379741 TSecr=2914295201 WS=128 |
| 3   | 0.000769  | 192.229.1.2 | 192.229.3.2 | TCP      | 66     | 37140 → 22 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2914295202 TSecr=3182379741                                |
| 4   | 0.000897  | 192.229.1.2 | 192.229.3.2 | TCP      | 66     | 37140 → 22 [FIN, ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2914295202 TSecr=3182379741                           |
| 5   | 0.001151  | 192.229.3.2 | 192.229.1.2 | TCP      | 66     | 22 → 37140 [FIN, ACK] Seq=1 Ack=2 Win=65280 Len=0 TSval=3182379741 TSecr=2914295202                           |
| 6   | 0.001682  | 192.229.1.2 | 192.229.3.2 | TCP      | 66     | 37140 → 22 [ACK] Seq=2 Ack=2 Win=64256 Len=0 TSval=2914295203 TSecr=3182379741                                |
| 11  | 15.075820 | 192.229.1.2 | 192.229.3.2 | TCP      | 74     | 38062 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2914310277 TSecr=0 WS=128                     |
| 12  | 15.076050 | 192.229.3.2 | 192.229.1.2 | TCP      | 74     | 80 → 38062 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=3182394816 TSecr=2914310277 WS=128 |
| 13  | 15.076489 | 192.229.1.2 | 192.229.3.2 | TCP      | 66     | 38062 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2914310278 TSecr=3182394816                                |
| 14  | 15.077036 | 192.229.1.2 | 192.229.3.2 | TCP      | 66     | 38062 → 80 [FIN, ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2914310278 TSecr=3182394816                           |
| 15  | 15.077299 | 192.229.3.2 | 192.229.1.2 | TCP      | 66     | 80 → 38062 [FIN, ACK] Seq=1 Ack=2 Win=65280 Len=0 TSval=3182394817 TSecr=2914310278                           |
| 16  | 15.077704 | 192.229.1.2 | 192.229.3.2 | TCP      | 66     | 38062 → 80 [ACK] Seq=2 Ack=2 Win=64256 Len=0 TSval=2914310279 TSecr=3182394817                                |
| 17  | 21.067638 | 192.229.1.2 | 192.229.3.2 | TCP      | 74     | 53460 → 7777 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2914316269 TSecr=0 WS=128                   |
| 18  | 21.067769 | 192.229.3.2 | 192.229.1.2 | TCP      | 54     | 7777 → 53460 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0                                                               |

**13 yaAllah**

Konsol Knights : apk update

apk add openssh

ssh-keygen -A

passwd root : jarkomwaput

CONFIGGGG nano /etc/ssh/sshd\_config :

PubkeyAuthentication yes

**PasswordAuthentication no**

PermitRootLogin yes

```
/usr/sbin/sshd
```

```
netstat -tuln | grep 22 (UDH LISTEN)
```

Konsol Mika diadminin : apk update

```
apk add openssh-client
```

```
apk add shadow
```

```
useradd -m mika_admin
```

```
su - mika_admin
```

```
ssh-keygen -t rsa -b 2048
```

```
cat /home/mika_admin/.ssh/id_rsa.pub
```

```
mkdir -p /root/.ssh
```

```
nano /root/.ssh/authorized_keys
```

```
chmod 700 /root/.ssh
```

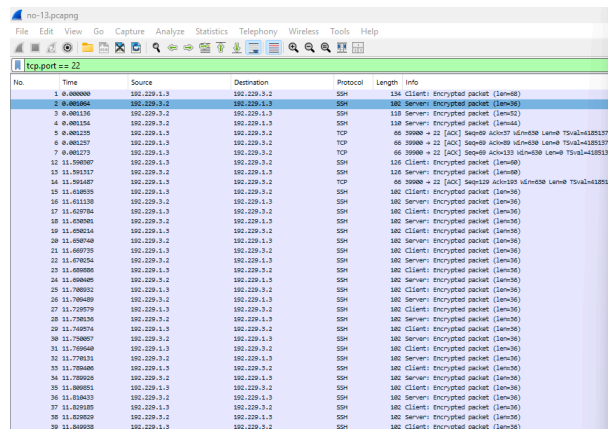
```
chmod 600 /root/.ssh/authorized_keys
```

```
pkill sshd
```

```
/usr/sbin/sshd
```

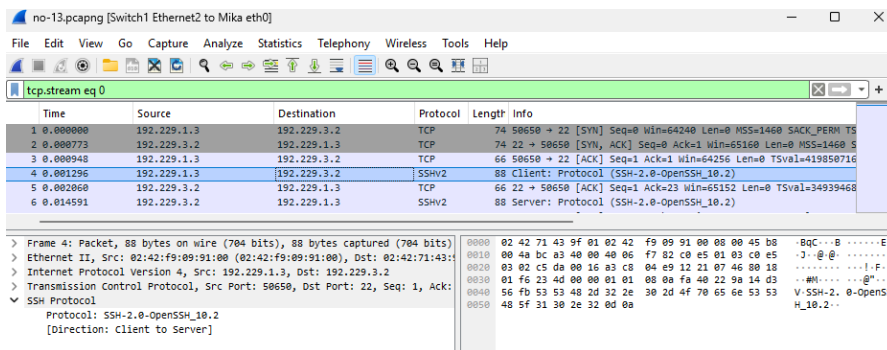
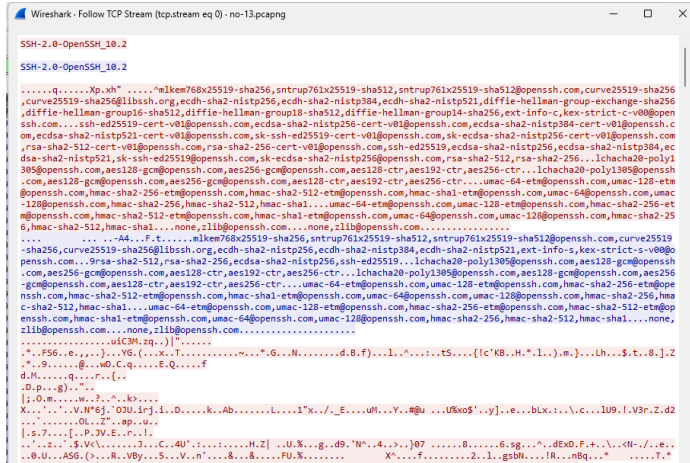
Capture ya

Ke mika admin : ssh root@<IP\_Knights>



The screenshot shows a Wireshark capture of network traffic on interface 'tcp.port == 22'. The packet list on the left shows 30 packets, all of which are SSH-related. The packet details pane on the right shows the structure of an SSH packet, including the 'SSH' protocol, 'Client: Encrypted packet (len=8)', and 'Server: Encrypted packet (len=8)'. The packet bytes pane at the bottom shows the raw data of the packet.

| No. | Time      | Source      | Destination | Protocol | Length | Info                                                       |
|-----|-----------|-------------|-------------|----------|--------|------------------------------------------------------------|
| 1   | 0.000000  | 192.168.1.1 | 192.168.1.2 | SSH      | 134    | Client: Encrypted packet (len=8)                           |
| 2   | 0.000004  | 192.168.1.2 | 192.168.1.1 | SSH      | 180    | Server: Encrypted packet (len=8)                           |
| 3   | 0.000116  | 192.168.1.2 | 192.168.1.3 | SSH      | 110    | Server: Encrypted packet (len=8)                           |
| 4   | 0.000116  | 192.168.1.2 | 192.168.1.3 | SSH      | 110    | Server: Encrypted packet (len=8)                           |
| 5   | 0.000125  | 192.168.1.3 | 192.168.1.2 | TCP      | 66     | 59080 -> 22 [ACK] Seq=89 Ack=37 Win=0 Len=0 TSval=43251705 |
| 6   | 0.000127  | 192.168.1.3 | 192.168.1.2 | TCP      | 66     | 59080 -> 22 [ACK] Seq=89 Ack=37 Win=0 Len=0 TSval=43251705 |
| 7   | 0.000173  | 192.168.1.3 | 192.168.1.2 | TCP      | 66     | 59080 -> 22 [ACK] Seq=89 Ack=37 Win=0 Len=0 TSval=43251705 |
| 12  | 11.590807 | 192.168.1.3 | 192.168.1.2 | SSH      | 110    | Client: Encrypted packet (len=8)                           |
| 13  | 11.591217 | 192.168.1.2 | 192.168.1.3 | SSH      | 110    | Server: Encrypted packet (len=8)                           |
| 14  | 11.591487 | 192.168.1.3 | 192.168.1.2 | TCP      | 66     | 59080 -> 22 [ACK] Seq=89 Ack=37 Win=0 Len=0 TSval=43251705 |
| 15  | 11.618035 | 192.168.1.3 | 192.168.1.2 | SSH      | 180    | Client: Encrypted packet (len=8)                           |
| 16  | 11.618138 | 192.168.1.2 | 192.168.1.3 | SSH      | 180    | Server: Encrypted packet (len=8)                           |
| 17  | 11.628784 | 192.168.1.3 | 192.168.1.2 | SSH      | 180    | Client: Encrypted packet (len=8)                           |
| 18  | 11.628881 | 192.168.1.2 | 192.168.1.3 | SSH      | 180    | Server: Encrypted packet (len=8)                           |
| 19  | 11.630034 | 192.168.1.3 | 192.168.1.2 | SSH      | 180    | Client: Encrypted packet (len=8)                           |
| 20  | 11.630148 | 192.168.1.2 | 192.168.1.3 | SSH      | 180    | Server: Encrypted packet (len=8)                           |
| 21  | 11.630175 | 192.168.1.3 | 192.168.1.2 | SSH      | 180    | Client: Encrypted packet (len=8)                           |
| 22  | 11.630254 | 192.168.1.2 | 192.168.1.3 | SSH      | 180    | Server: Encrypted packet (len=8)                           |
| 23  | 11.630306 | 192.168.1.3 | 192.168.1.2 | SSH      | 180    | Client: Encrypted packet (len=8)                           |
| 24  | 11.630400 | 192.168.1.2 | 192.168.1.3 | SSH      | 180    | Server: Encrypted packet (len=8)                           |
| 25  | 11.708051 | 192.168.1.3 | 192.168.1.2 | SSH      | 180    | Client: Encrypted packet (len=8)                           |
| 26  | 11.708060 | 192.168.1.2 | 192.168.1.3 | SSH      | 180    | Server: Encrypted packet (len=8)                           |
| 27  | 11.725979 | 192.168.1.3 | 192.168.1.2 | SSH      | 180    | Client: Encrypted packet (len=8)                           |
| 28  | 11.726030 | 192.168.1.2 | 192.168.1.3 | SSH      | 180    | Server: Encrypted packet (len=8)                           |
| 29  | 11.726074 | 192.168.1.3 | 192.168.1.2 | SSH      | 180    | Client: Encrypted packet (len=8)                           |
| 30  | 11.726087 | 192.168.1.2 | 192.168.1.3 | SSH      | 180    | Server: Encrypted packet (len=8)                           |
| 31  | 11.726098 | 192.168.1.3 | 192.168.1.2 | SSH      | 180    | Client: Encrypted packet (len=8)                           |
| 32  | 11.726131 | 192.168.1.2 | 192.168.1.3 | SSH      | 180    | Server: Encrypted packet (len=8)                           |
| 33  | 11.726148 | 192.168.1.3 | 192.168.1.2 | SSH      | 180    | Client: Encrypted packet (len=8)                           |
| 34  | 11.726158 | 192.168.1.2 | 192.168.1.3 | SSH      | 180    | Server: Encrypted packet (len=8)                           |
| 35  | 11.800051 | 192.168.1.3 | 192.168.1.2 | SSH      | 180    | Client: Encrypted packet (len=8)                           |
| 36  | 11.820453 | 192.168.1.2 | 192.168.1.3 | SSH      | 180    | Server: Encrypted packet (len=8)                           |
| 37  | 11.820500 | 192.168.1.3 | 192.168.1.2 | SSH      | 180    | Client: Encrypted packet (len=8)                           |
| 38  | 11.820509 | 192.168.1.2 | 192.168.1.3 | SSH      | 180    | Server: Encrypted packet (len=8)                           |
| 39  | 11.820518 | 192.168.1.3 | 192.168.1.2 | SSH      | 180    | Client: Encrypted packet (len=8)                           |



CONFIGGGG : nano /root/setup\_ssh.sh

#!/bin/sh

apk update

apk add openssh

# generate host key kalau belum ada

[ -f /etc/ssh/ssh\_host\_rsa\_key ] || ssh-keygen -A

# jalankan sshd

kill sshd 2>/dev/null

/usr/sbin/sshd

Lalu

```
chmod +x /root/setup_ssh.sh
mkdir -p /root/backup_conf
cp /etc/ssh/sshd_config /root/backup_conf/sshd_config
cp /root/backup_conf/sshd_config /etc/ssh/sshd_config
pkill sshd
/usr/sbin/sshd
```

#### 14 HAHHAHHA

| Item                   | Nilai             |
|------------------------|-------------------|
| IP Penyerang           | 172.26.7.50       |
| IP Target              | 172.26.7.100      |
| Port yang diserang     | 8080              |
| Endpoint               | /login.php        |
| Username               | lain_admin        |
| Password yang berhasil | wired_pr0tocol_7  |
| Server                 | Apache/2.4.62\r\n |

```
alice X chisa X eiri X knight X miika X nazwa X
nazwaadp@DESKTOP-RQEOF0P: ~$ nc 10.4.89.247 3401

===== Soal 14 - Protocol 7: Brute Force Analysis =====
Difficulty: Easy
Note: You can exit anytime by typing 'exit'

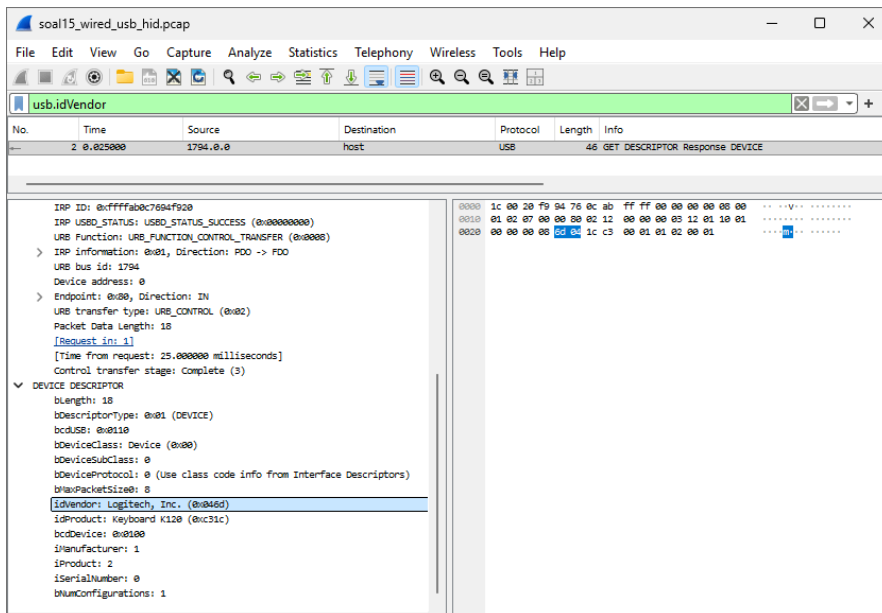
What is the IP address of the attacker performing the brute force attack?
Format: IP
> 172.26.7.50

What is the target IP and port being attacked?
Format: IP:port
> 172.26.7.100:8080

What is the password found for the user lain_admin?
Format: string
> wired_protocol_7

What is the web server software and version reported in the response header?
Format: Software/X.X.X (e.g. Apache/2.4.0)
> Apache/2.4.62
Congratulations! Here is your flag: KOMJAR26{Wlr3d_Brut3_VgPZlNqRRrWds9C9DilAdql1}
```

15 HAHHAHHA



Temuan :

idVendor : 0x046d

idProduct : 0xc31c



Temuan : Wired\_Protocol\_7\_is\_alive\_2026

```
nazwaadp@DESKTOP-RQE0F0P: X + v
nazwaadp@DESKTOP-RQE0F0P:~$ nc 10.4.89.246 3402

===== Soal 15 - Protocol 7: USB Keystroke Decoding =====
Difficulty: Medium
Note: You can exit anytime by typing 'exit'

What is the Vendor ID of the captured USB HID device?
Format: 0x1234 (or 1234)
> 0x046d

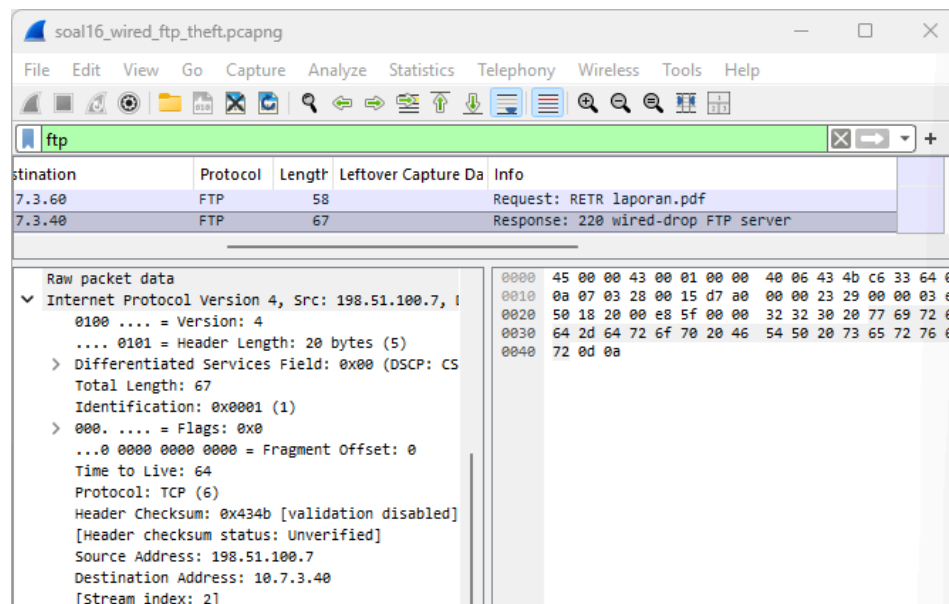
What is the Product ID of the captured USB HID device?
Format: 0x1234 (or 1234)
> 0xc31c

What is the USB device address assigned to the keyboard?
Format: int
> 7

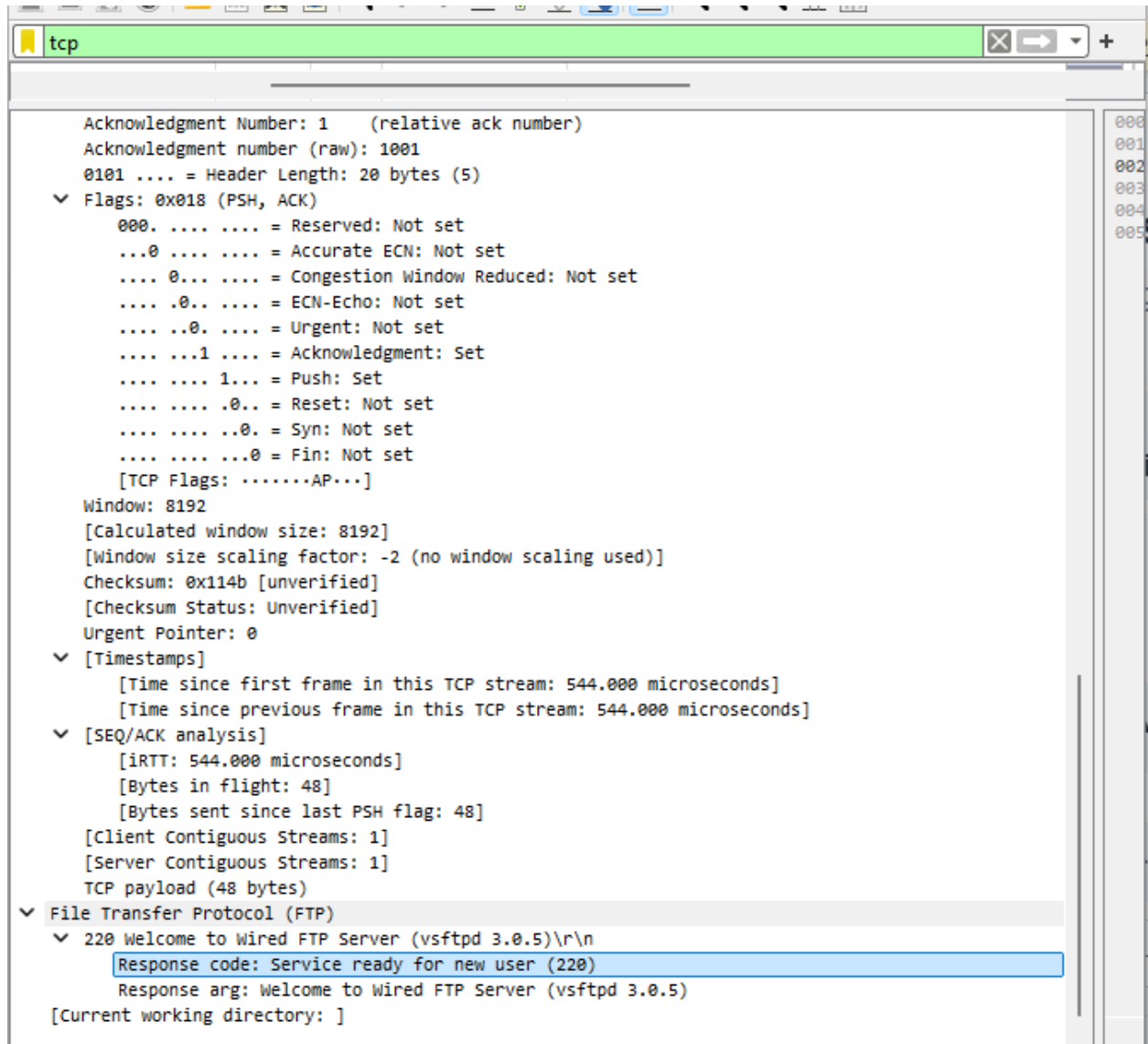
What is the secret message decoded from the captured keystrokes?
Format: string
> Wired_Protocol_7_is_alive_2026
Congratulations! Here is your flag: KOMJAR26{USB_K3ystr0k3_LQST4m7Lh0KPxoEWMvgJ9Ui07y}
```

Validasi:

## 16 HAHHAHHA



Temuan : IP SOURCE 198.51.100.7



Temuan : Wired FTP Server (vsftpd 3.0.5)

```
Wireshark - Follow TCP Stream (tcp.stream eq 6) - soal16_wired_ftp_theft.pcapng

220 Welcome to Wired FTP Server (vsftpd 3.0.5)
USER knights_agent
331 Please specify the password.
PASS N4v1_s3cur3_2026
230 Login successful.
PWD
257 "/" is the current directory
TYPE I
200 Switching to Binary mode.
SIZE knights_payload.exe
213 524288
PASV
227 Entering Passive Mode (198,51,100,7,156,64).
RETR knights_payload.exe
150 Opening BINARY mode data connection for knights_payload.exe (524288 bytes).
226 Transfer complete.
QUIT
221 Goodbye.
```

Temuan :

User : knights\_agent

Pass : N4v1\_s3cur3\_2026

Size (bytes) : 524288

```
===== Soal 16 - Protocol 7: FTP Credential Theft =====
Difficulty: Hard
Note: You can exit anytime by typing 'exit'

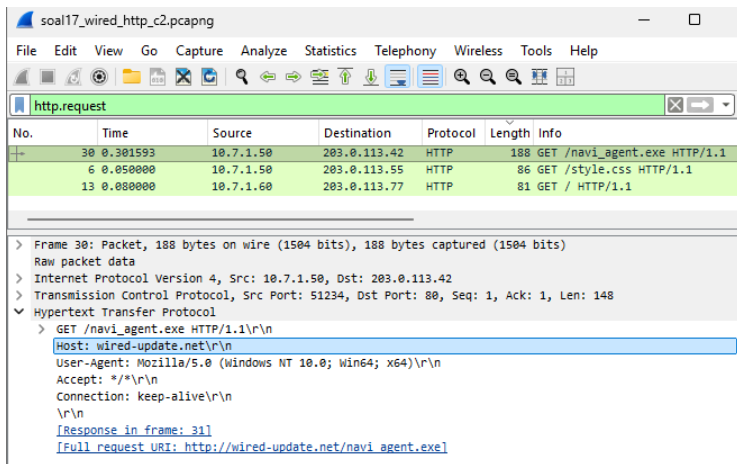
What is the IP address of the FTP server used to download the malware?
Format: IP
> 198.51.100.7

What FTP server software banner is returned upon connection?
Format: string
> Wired FTP Server (vsftpd 3.0.5)

What credential did the attacker use to log in to the FTP server?
Format: user:pass
> knights_agent:N4v1_s3cur3_2026

What is the size in bytes of the malware file (knights_payload.exe) requested via FTP?
Format: int
> 524288
Congratulations! Here is your flag: KOMJAR26{FTP_Th3ft_vZKUn7nugxuRzmThvRZUuV4wT}
```

17 wkwk



soal17\_wired\_http\_c2.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http.request

| No. | Time     | Source    | Destination  | Protocol | Length | Info                         |
|-----|----------|-----------|--------------|----------|--------|------------------------------|
| 30  | 0.301593 | 10.7.1.50 | 203.0.113.42 | HTTP     | 188    | GET /navi_agent.exe HTTP/1.1 |
| 6   | 0.050000 | 10.7.1.50 | 203.0.113.55 | HTTP     | 86     | GET /style.css HTTP/1.1      |
| 13  | 0.080000 | 10.7.1.60 | 203.0.113.77 | HTTP     | 81     | GET / HTTP/1.1               |

> Frame 30: Packet, 188 bytes on wire (1504 bits), 188 bytes captured (1504 bits)

Raw packet data

> Internet Protocol Version 4, Src: 10.7.1.50, Dst: 203.0.113.42

> Transmission Control Protocol, Src Port: 51234, Dst Port: 80, Seq: 1, Ack: 1, Len: 148

> Hypertext Transfer Protocol

> GET /navi\_agent.exe HTTP/1.1\r\n

Host: wired-update.net\r\n

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64)\r\n

Accept: \*/\*\r\n

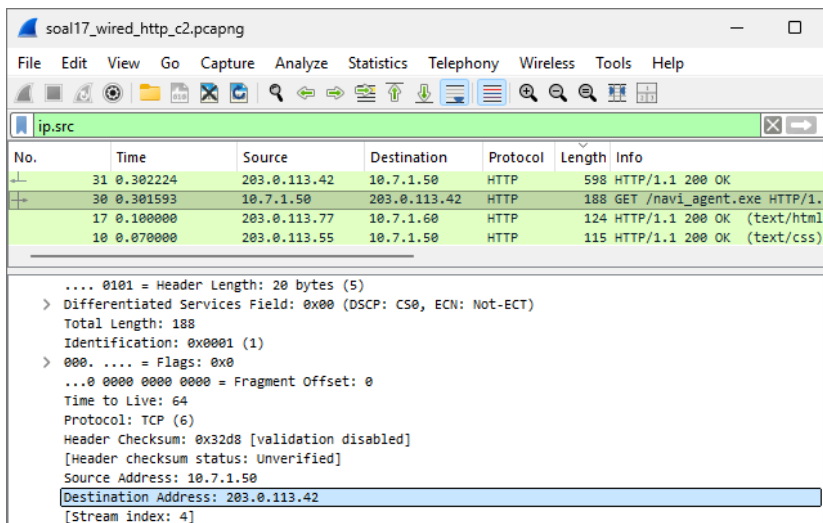
Connection: keep-alive\r\n

\r\n

[Response in frame: 31]

[Full request URI: http://wired-update.net/navi\_agent.exe]

domain (Host) : wired-update.net



soal17\_wired\_http\_c2.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ip.src

| No. | Time     | Source       | Destination  | Protocol | Length | Info                         |
|-----|----------|--------------|--------------|----------|--------|------------------------------|
| 31  | 0.302224 | 203.0.113.42 | 10.7.1.50    | HTTP     | 598    | HTTP/1.1 200 OK              |
| 30  | 0.301593 | 10.7.1.50    | 203.0.113.42 | HTTP     | 188    | GET /navi_agent.exe HTTP/1.1 |
| 17  | 0.100000 | 203.0.113.77 | 10.7.1.60    | HTTP     | 124    | HTTP/1.1 200 OK (text/html)  |
| 10  | 0.070000 | 203.0.113.55 | 10.7.1.50    | HTTP     | 115    | HTTP/1.1 200 OK (text/css)   |

.... 0101 = Header Length: 20 bytes (5)

> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 188

Identification: 0x0001 (1)

> 000. .... = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 64

Protocol: TCP (6)

Header Checksum: 0x32d8 [validation disabled]

[Header checksum status: Unverified]

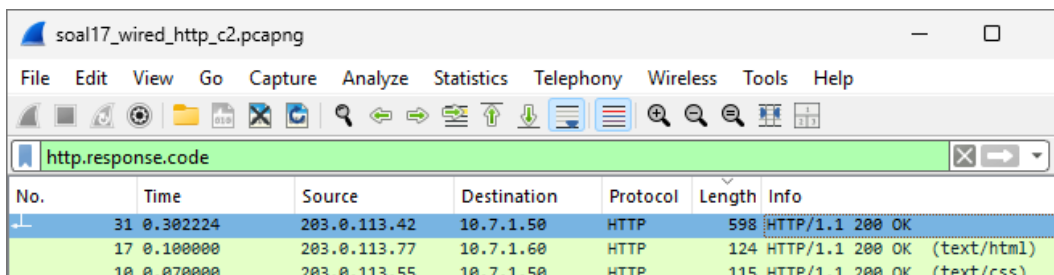
Source Address: 10.7.1.50

Destination Address: 203.0.113.42

[Stream index: 4]

IP server penyerang : 203.0.113.42

nama file exe : navi\_agent.exe



soal17\_wired\_http\_c2.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http.response.code

| No. | Time     | Source       | Destination | Protocol | Length | Info                        |
|-----|----------|--------------|-------------|----------|--------|-----------------------------|
| 31  | 0.302224 | 203.0.113.42 | 10.7.1.50   | HTTP     | 598    | HTTP/1.1 200 OK             |
| 17  | 0.100000 | 203.0.113.77 | 10.7.1.60   | HTTP     | 124    | HTTP/1.1 200 OK (text/html) |
| 10  | 0.070000 | 203.0.113.55 | 10.7.1.50   | HTTP     | 115    | HTTP/1.1 200 OK (text/css)  |

kode status HTTP : 200

## 18 buosss

soal18\_wired\_smb\_transfer.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wi

smb2

| No. | Time     | Source     | Destination | Protocol |
|-----|----------|------------|-------------|----------|
| 4   | 0.001881 | 10.7.3.100 | 10.7.1.50   | SMB2     |
| 6   | 0.002985 | 10.7.1.50  | 10.7.3.100  | SMB2     |
| 8   | 0.004085 | 10.7.3.100 | 10.7.1.50   | SMB2     |
| 10  | 0.004732 | 10.7.1.50  | 10.7.3.100  | SMB2     |
| 12  | 0.005330 | 10.7.3.100 | 10.7.1.50   | SMB2     |
| 14  | 0.005330 | 10.7.1.50  | 10.7.3.100  | SMB2     |
| 16  | 0.006049 | 10.7.3.100 | 10.7.1.50   | SMB2     |
| 18  | 0.006896 | 10.7.1.50  | 10.7.3.100  | SMB2     |
| 20  | 0.007744 | 10.7.3.100 | 10.7.1.50   | SMB2     |
| 22  | 0.008575 | 10.7.1.50  | 10.7.3.100  | SMB2     |
| 24  | 0.009352 | 10.7.3.100 | 10.7.1.50   | SMB2     |
| 26  | 0.009352 | 10.7.1.50  | 10.7.3.100  | SMB2     |

nama protokol : smb2

soal18\_wired\_smb\_transfer.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

smb2

| No. | Time     | Source     | Destination | Protocol | Length | Info                          |
|-----|----------|------------|-------------|----------|--------|-------------------------------|
| 4   | 0.001881 | 10.7.3.100 | 10.7.1.50   | SMB2     | 146    | Negotiate Protocol Request    |
| 6   | 0.002985 | 10.7.1.50  | 10.7.3.100  | SMB2     | 172    | Negotiate Protocol Response   |
| 8   | 0.004085 | 10.7.3.100 | 10.7.1.50   | SMB2     | 132    | Session Setup Request         |
| 10  | 0.004732 | 10.7.1.50  | 10.7.3.100  | SMB2     | 116    | Session Setup Response        |
| 12  | 0.005330 | 10.7.3.100 | 10.7.1.50   | SMB2     | 152    | Tree Connect Request, Tree: ' |
| 14  | 0.005330 | 10.7.1.50  | 10.7.3.100  | SMB2     | 124    | Tree Connect Response, Tree:  |
| 16  | 0.006049 | 10.7.3.100 | 10.7.1.50   | SMB2     | 230    | Create Request, File: System3 |
| 18  | 0.006896 | 10.7.1.50  | 10.7.3.100  | SMB2     | 196    | Create Response, File: System |
| 20  | 0.007744 | 10.7.3.100 | 10.7.1.50   | SMB2     | 1184   | Write Request Len:1028 Off:0, |
| 22  | 0.008575 | 10.7.1.50  | 10.7.3.100  | SMB2     | 124    | Write Response, File: System3 |
| 24  | 0.009352 | 10.7.3.100 | 10.7.1.50   | SMB2     | 132    | Close Request, File: System32 |
| 26  | 0.009352 | 10.7.1.50  | 10.7.3.100  | SMB2     | 168    | Close Response, File: System3 |

> Frame 4: Packet, 146 bytes on wire (1168 bits), 146 bytes captured (1168 bits)  
Raw packet data  
▼ Internet Protocol Version 4, Src: 10.7.3.100, Dst: 10.7.1.50

IP pengirim & penerima : 10.7.3.100 & 10.7.1.50

soal18\_wired\_smb\_transfer.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

smb2.filenamee

| Destination | Protocol | Length | Info                                                                  |
|-------------|----------|--------|-----------------------------------------------------------------------|
| 10.7.1.50   | SMB2     | 230    | Create Request, File: System32\wired_trojan_payload.exe               |
| 10.7.3.100  | SMB2     | 196    | Create Response, File: System32\wired_trojan_payload.exe              |
| 10.7.1.50   | SMB2     | 1184   | Write Request Len:1028 Off:0, File: System32\wired_trojan_payload.exe |
| 10.7.3.100  | SMB2     | 124    | Write Response, File: System32\wired_trojan_payload.exe               |
| 10.7.1.50   | SMB2     | 132    | Close Request, File: System32\wired_trojan_payload.exe                |
| 10.7.3.100  | SMB2     | 168    | Close Response, File: System32\wired_trojan_payload.exe               |

folder tujuan : system32

nama file exe. : wired\_trojan\_payload.exe

```
nazwaadp@DESKTOP-RQE0F0P:~$ nc 10.4.89.246 3405

===== Soal 18 - Protocol 7: SMB Lateral Transfer =====
Difficulty: Hard
Note: You can exit anytime by typing 'exit'

What network file sharing protocol was used to transfer the malware to the victim?
Format: string
> smb2

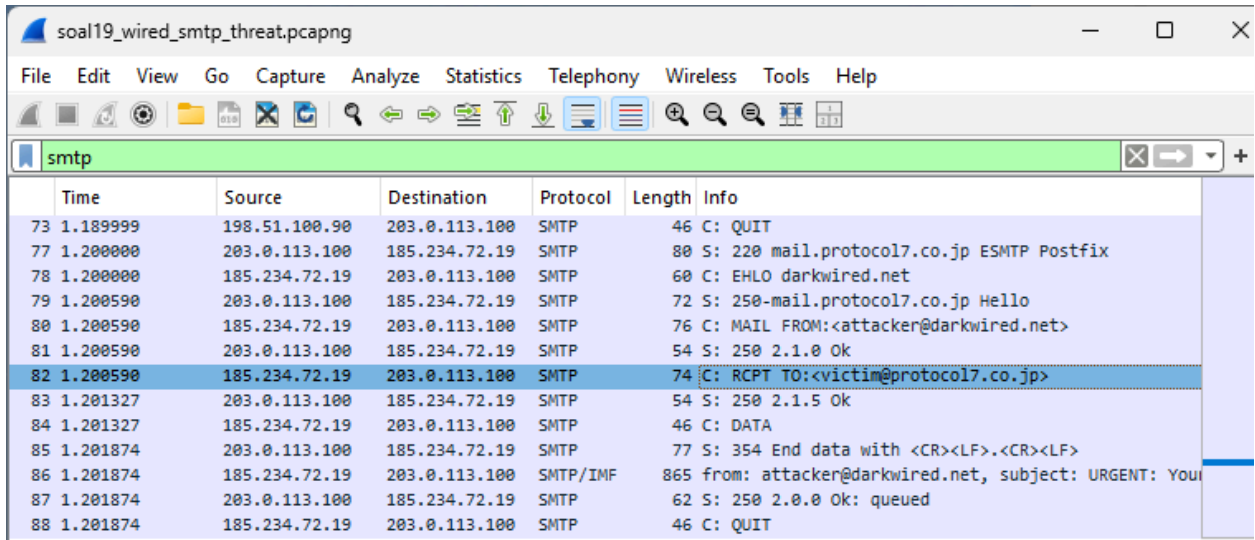
What is the IP address of the source host delivering the malware?
Format: IP
> 10.7.3.100

What is the IP address of the victim host receiving the malware?
Format: IP
> 10.7.1.50

What target share or directory on the victim was the malware written to?
Format: string
> system32

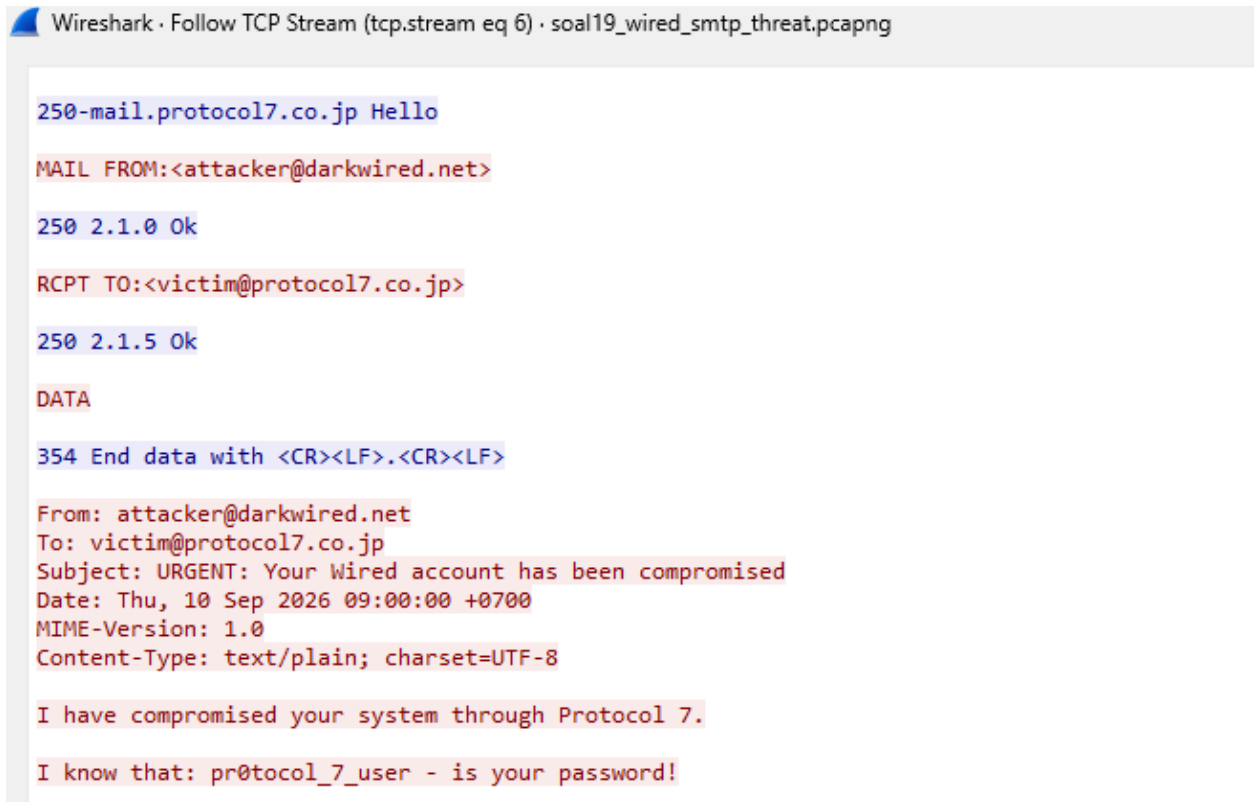
What is the filename of the executable malware transferred?
Format: file.exe
> wired_trojan_payload.exe
Congratulations! Here is your flag: KOMJAR26{SMB_Tr4nsf3r_ay9WxR5WJSmzpdVLSg2QfDnYh}
```

## NO 19 SUMPAHHHH



| Time        | Source        | Destination   | Protocol | Length | Info                                                |
|-------------|---------------|---------------|----------|--------|-----------------------------------------------------|
| 73 1.189999 | 198.51.100.90 | 203.0.113.100 | SMTP     | 46     | C: QUIT                                             |
| 77 1.200000 | 203.0.113.100 | 185.234.72.19 | SMTP     | 80     | S: 220 mail.protocol7.co.jp ESMTP Postfix           |
| 78 1.200000 | 185.234.72.19 | 203.0.113.100 | SMTP     | 60     | C: EHLO darkwired.net                               |
| 79 1.200590 | 203.0.113.100 | 185.234.72.19 | SMTP     | 72     | S: 250-mail.protocol7.co.jp Hello                   |
| 80 1.200590 | 185.234.72.19 | 203.0.113.100 | SMTP     | 76     | C: MAIL FROM:<attacker@darkwired.net>               |
| 81 1.200590 | 203.0.113.100 | 185.234.72.19 | SMTP     | 54     | S: 250 2.1.0 Ok                                     |
| 82 1.200590 | 185.234.72.19 | 203.0.113.100 | SMTP     | 74     | C: RCPT TO:<victim@protocol7.co.jp>                 |
| 83 1.201327 | 203.0.113.100 | 185.234.72.19 | SMTP     | 54     | S: 250 2.1.5 Ok                                     |
| 84 1.201327 | 185.234.72.19 | 203.0.113.100 | SMTP     | 46     | C: DATA                                             |
| 85 1.201874 | 203.0.113.100 | 185.234.72.19 | SMTP     | 77     | S: 354 End data with <CR><LF>.<CR><LF>              |
| 86 1.201874 | 185.234.72.19 | 203.0.113.100 | SMTP/IMF | 865    | from: attacker@darkwired.net, subject: URGENT: Your |
| 87 1.201874 | 203.0.113.100 | 185.234.72.19 | SMTP     | 62     | S: 250 2.0.0 Ok: queued                             |
| 88 1.201874 | 185.234.72.19 | 203.0.113.100 | SMTP     | 46     | C: QUIT                                             |

email korban : victim@protocol7.co.jp



250-mail.protocol7.co.jp Hello

MAIL FROM:<attacker@darkwired.net>

250 2.1.0 Ok

RCPT TO:<victim@protocol7.co.jp>

250 2.1.5 Ok

DATA

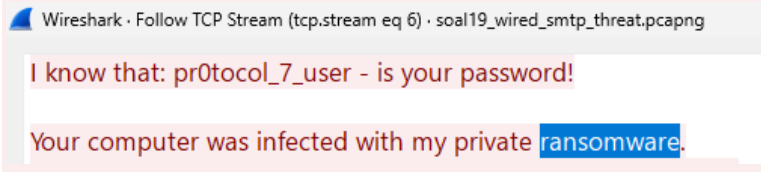
354 End data with <CR><LF>.<CR><LF>

From: attacker@darkwired.net  
To: victim@protocol7.co.jp  
Subject: URGENT: Your Wired account has been compromised  
Date: Thu, 10 Sep 2026 09:00:00 +0700  
MIME-Version: 1.0  
Content-Type: text/plain; charset=UTF-8

I have compromised your system through Protocol 7.

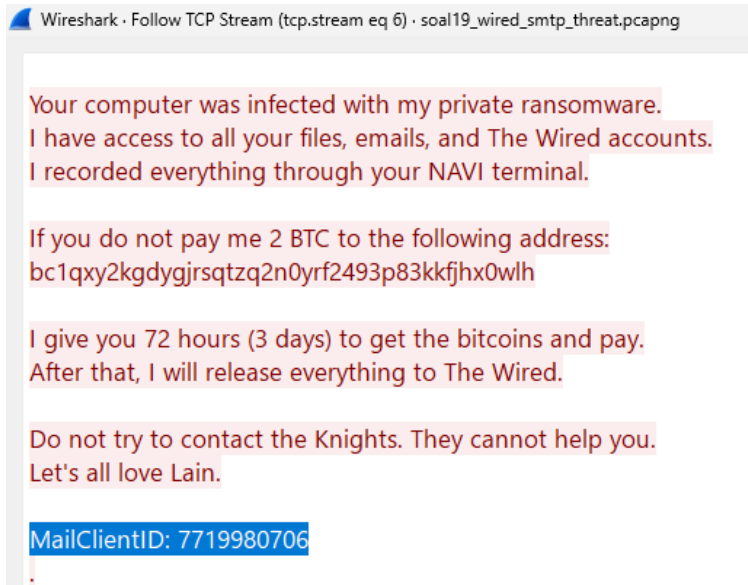
I know that: pr0tocol\_7\_user - is your password!

password yang diklaim bocor : pr0tocol\_7\_user

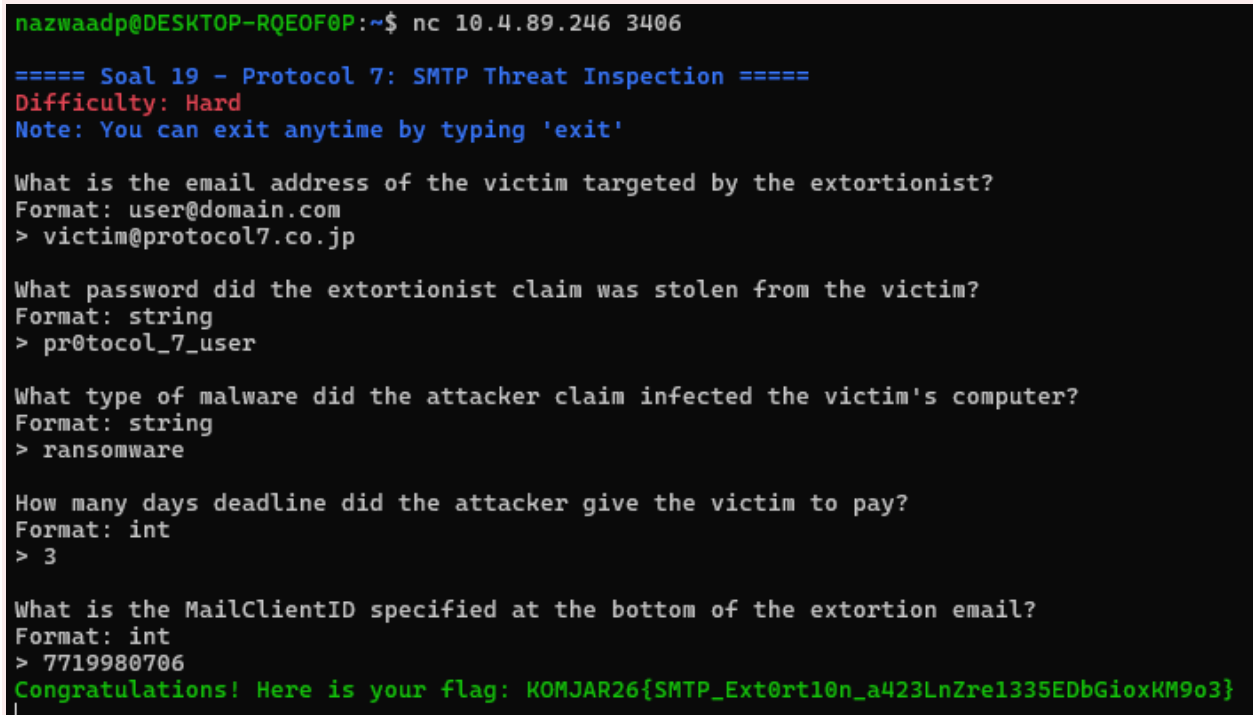


jenis malware : ransomware

deadline (hari) : I give you 72 hours (3 days)

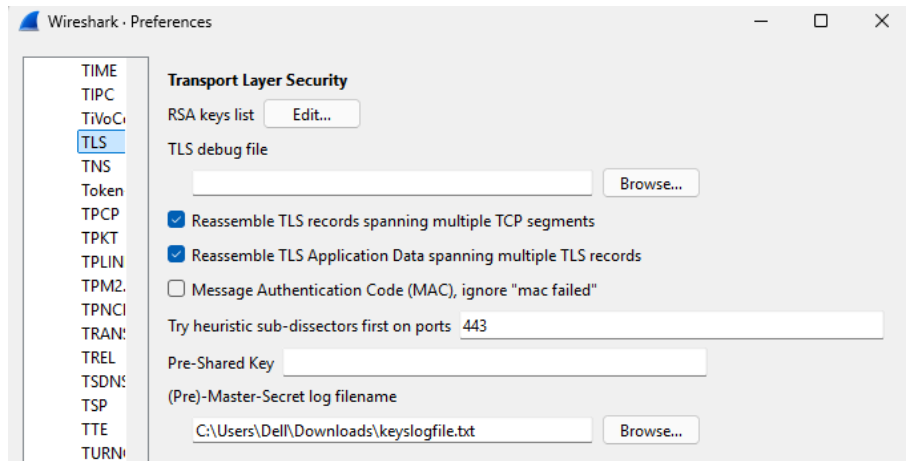


MailClientID.: 7719980706



## 20 ALHAMDULILLAHHHHHH

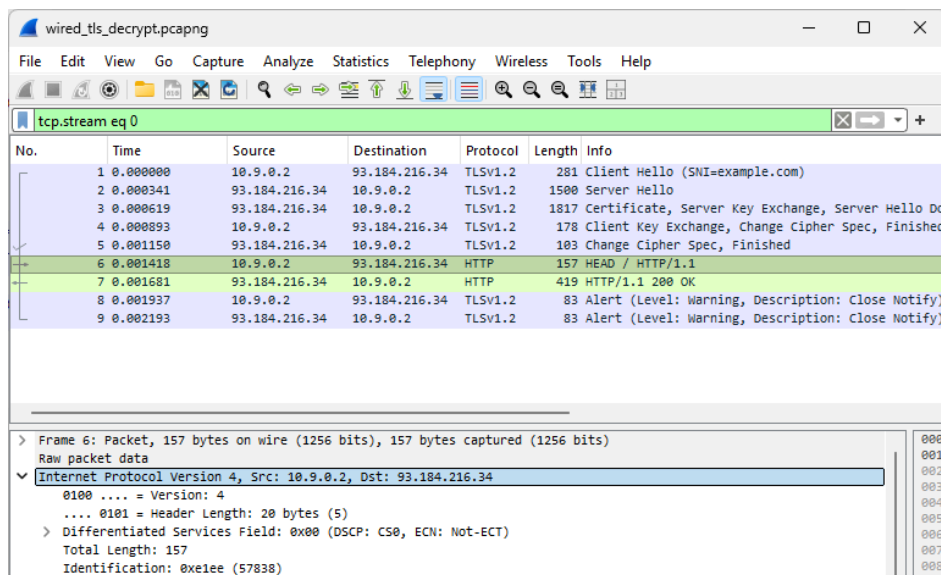
"(Pre)-Master-Secret log filename", arahkan ke keyslogfile.txt



| No. | Time     | Source   | Destination   | Protocol | Length | Info                           |
|-----|----------|----------|---------------|----------|--------|--------------------------------|
| 1   | 0.000000 | 10.9.0.2 | 93.184.216.34 | TLSv1.2  | 281    | Client Hello (SNI=example.com) |

versi TLS : TLSv1.2

SNI domain : example.com



IP server HTTPS penyerang : 93.184.216.34

Wireshark · Follow HTTP Stream (tcp.stream eq 0) · wired\_tls\_decrypt.pcapng

```
HEAD / HTTP/1.1
Host: example.com
User-Agent: curl/7.62.0
Accept: */*
```

User-Agent : curl/7.62.0

Wireshark · Follow HTTP Stream (tcp.stream eq 0) · wired\_tls\_decrypt.pcapng

```
HEAD / HTTP/1.1
Host: example.com
User-Agent: curl/7.62.0
Accept: */*
```

HTTP method+path tersembunyi. : HEAD / HTTP/1.1

```
nazwaadp@DESKTOP-RQEOF0P:~$ nc 10.4.89.246 3407
```

```
==== Soal 20 - Protocol 7: TLS Decrypted Stream ====
```

```
Difficulty: Hard
```

```
Note: You can exit anytime by typing 'exit'
```

```
What specific TLS protocol version was negotiated for the encrypted communication?
```

```
Format: string (e.g. TLSv1.2)
```

```
> TLSv1.2
```

```
What domain name (SNI / Host) was requested by the client during the TLS handshake?
```

```
Format: domain.com
```

```
> example.com
```

```
What is the IP address of the HTTPS server?
```

```
Format: IP
```

```
> 93.184.216.34
```

```
What User-Agent string was used by the client during the decrypted HTTP session?
```

```
Format: string
```

```
> curl/7.62.0
```

```
What HTTP request method and path was sent in the decrypted request?
```

```
Format: METHOD /path (e.g. GET /index.html)
```

```
> HEAD / HTTP/1.1
```

```
Congratulations! Here is your flag: KOMJAR26{TLS_D3crypt_oaTBwgD0IH9wkYU6ecvnYypSJ}
```

# KELAARRRRRRRRRRRRRRRR.